

Supplemental Material: Table of Excluded Studies (Full-Text Stage)

Overview

Following a comprehensive full-text eligibility assessment of 117 candidate studies, a total of 110 studies met the inclusion criteria for this systematic literature review. The remaining seven studies were excluded from the final synthesis. This table provides a transparent record of those exclusions, categorized by their failure to meet specific technical requirements regarding **Preprocessing, feature engineering for human behavioral insider threat detection (ITD)** or failing to meet the minimum **Quality Assessment (QA)** thresholds.

#	Author (Year)	Paper Title	Reason for Exclusion
1	Agarkar (2025)	Assessing the impact of user behavior and insider threats on critical infrastructure	Lacks technical detail and specific algorithms for the data preprocessing stage (cleaning, normalization).
2	Ahmad (2021)	Mitigating malicious insider attacks in the internet of things using Supervised machine learning techniques	Utilizes the NSL-KDD network dataset; reflects low-level network traffic instead of organizational user behavior.
3	Alnaim (2025)	Zero-Trust Mechanisms for Securing Distributed Edge and Fog Computing in 6G Networks	Focuses on network traffic/6G hardware trust rather than human behavioral feature engineering.
4	Kandanaarachchi (2021)	Honeyboost: Boosting honeypot performance with data fusion and anomaly detection	Focused on network protocols and honeypots without expert-defined human behavioral indicators.
5	Mahmoudi-Nasr (2024)	An authorization framework to mitigate insider threat in CIM-based smart grid	Utilizes manual statistical formulas (EWMA) rather than ML/DL-based feature engineering frameworks.
6	Sulaiman et al. (2023)	Quantitative Analysis of Worm Transmission and Insider Risks in Air-Gapped Networking Using a Novel Machine Learning Approach	Uses ML as a mathematical solver for worm transmission equations rather than an ITD detection framework.
7	Yusof et al. (2022)	Behavioral Intrusion Prediction Model on Bayesian Network over Healthcare Infrastructure	Relies strictly on network-layer features (PCAP) and lacks human/expert behavioral rules.

Table 2: Detailed Quality Assessment (QA) Scoring for Excluded Studies

he following papers were evaluated against five Quality Assessment criteria (QA1–QA5). Each criterion is scored out of 1.0, with a maximum possible score of 5.0.

QA Criteria Legend:

- **QA1:** Dataset Validity (Specific to ITD)
- **QA2:** Preprocessing Rigor (Cleaning/Normalization)
- **QA3:** Feature Engineering/Expert Indicators
- **QA4:** Model Architecture Depth
- **QA5:** Performance Metrics & Evaluation

#	Author (Year)	QA1: Dataset (Justification)	QA2: Preprocessing (Justification)	QA3: Expert Indicators (Justification)	QA4: Automated Architecture (Justification)	QA5: Results (Justification)	Total	Decision
1	Agarkar (2025) Assessing the impact of user behavior and insider threats on critical infrastructure	1.0: Identifies use of anonymized logs , incident reports, and interviews from critical infrastructure sectors.	0.0: Mentions "Preprocessing and Data Cleaning" as a stage but does not provide technical algorithms for cleaning, noise reduction, or normalization.	0.0: Mentions " Expert Interviews " as a source but does not explicitly list the resulting indicators or rules used for the baseline.	1.0: Explains a framework using ML clustering (K-means, DBSCAN) and LSTM neural networks .	1.0: Reports a 92% TPR , 10% FPR, and 30% reduction in detection time.	3.0	Exclude
2	Ahmad (2021)	0.5: Uses the NSL-KDD network intrusion dataset. While	0.0: Describes standard normalization but lacks specific "User	0.5: The paper relies on raw network flow features (TCP/UDP); it	1.0: Explains supervised ensemble methods and SGD-based	1.0: Reports standard metrics (Accuracy, F1, AUC) for 15	3.0	Exclude

		identified, it reflects low-level network traffic rather than the organizational user behavior required by IC-1.	Behavior" cleaning (e.g., session reconstruction or identity mapping) essential for this SLR.	does not define any expert human-behavioral rules or manual indicators. (Sec 5.1)	learners for anomaly detection. (Sec 4)	algorithm variations. (Table 5.2)		
3	Alnaim (2025) Zero-Trust Mechanisms for Securing Distributed Edge and Fog Computing in 6G Networks	0.5: Uses a hybrid network dataset (6G Synthetic + UNSW-NB15) centered on network traffic rather than organizational UBA logs.	0.5: Describes general network normalization but lacks specific behavioral feature engineering (email, file access) required for your ITD scope.	0.0: Trust scores are based on hardware/network entities; lacks exclusive focus on human behavioral rules/indicators.	1.0: Strong explanation of a 12-layer AI and Blockchain-based framework for decentralized security.	1.0: Reports clear performance metrics for 6G latency, throughput, and general attack mitigation.	3.0	EXCLUDE
4	Kandanaarachchi (2021) Honeyboost: Boosting honeypot performance with data fusion and anomaly detection	0.5: Uses data from the "LAN Security Monitoring Project." While real-world, it is not a specialized ITD benchmark like CERT or Enron.	0.5: Focuses on network protocol packets (ARP, TCP, UDP) rather than organizational user-log sequences.	0.0: Zero expert-defined behavioral or psychological indicators for human insiders are utilized.	0.5: The "Honeyboost" framework is designed for honeypot enhancement and general Network Anomaly Detection (NADS), not specifically for insider behavior.	0.5: Detects "anomalous nodes," but lacks ground-truth verification of actual insider maliciousness (e.g., sabotage, theft).	2.0	Exclude
5	Mahmoudi-Nasr (2024) An authorization framework to mitigate insider threat in CIM-based smart grid	0.0: Uses a private simulation (UMZ ICS Lab) based on Iranian power system logs. No public ITD benchmark (CERT/Enron).	0.5: Describes CIM mapping, but lacks standard ML preprocessing like scaling or encoding for model training.	1.0: Identifies specific grid-based indicators (pending/cleared alarms). (Section 4)	0.0: Uses manual formulas (EWMA) and statistical control charts (I-MR). Does not use ML/DL architectures.	0.5: Results focus on access authorization success rather than standard ITD metrics like AUC or F1-score.	2.0	Exclude

6	Sulaiman et al. (2023) Quantitative Analysis of Worm Transmission and Insider Risks in Air-Gapped Networking Using a Novel Machine Learning Approach	0.0: Does not use behavioral logs (e.g., CERT r4.2). It uses numerical scenarios to solve Ordinary Differential Equation.	0.0: No behavior-based preprocessing; focuses on mathematical state initialization.	0.0: No behavioral indicators; uses epidemic parameters like infection and patch removal rates.	0.0: Uses ML (ANN) as a "surrogate solver" for math, not as a detection/classification framework.	0.5: Reports MSE and convergence for math solutions, not detection metrics like F1 or Precision for identifying users.	0.5	Exclude
7	Yusof et al. (2022) Behavioral Intrusion Prediction Model on Bayesian Network over Healthcare Infrastructure	1.0: Uses ground-truth production traffic (PCAP files) from a major Malaysian healthcare provider and simulated attacks. (Sec 3.1)	0.5: Defines a Feature Selection Model based on distribution density functions (Beta, Gamma, MLE), but lacks standard ML scaling/encoding details. (Sec 4.1)	0.0: Features are purely network-layer (window size, frame length, delta time) and do not include human/expert behavioral rules for insiders. (Tab 3)	0.5: Bayesian Network implementation used for probabilistic modeling and predictive analytics of Network Traffic. (Sec 3)	1.0: Presents Accuracy, False Positive Rate, and error measurements (AIC/BIC) compared against SVM and k-NN. (Sec 4.1)	3.0	Exclude

The following (7) records were excluded as they were not downloadable in full-text.

1- Insider threat detection and prevention using semantic score and dynamic multi-fuzzy classifier Malvika Singh , S. Sangeetha and B.M. Mehtre 2023

2- Explainable AI (XAI) for Insider Threat Detection: Balancing Security and Transparency in Cloud Computing Arul Selvam P, Tamije Selvy P 2025

3- User behaviour based insider threat detection model using an LSTM integrated RF model
S. K. Uma Maheswaran,L. Rajasekar,Ziaul Haque Choudhury &Makarand Shahade 2023

4- Identification of authorship and prevention of fraudulent transactions/cybercrime using efficient high performance machine learning techniques
B.J. Sowmya, R. Hanumantharaju, D. Pradeep Kumar and K.G. Srinivasa 2022

5- Deep Temporal Graph Infomax for Imbalanced Insider Threat Detection

Peng Gao Icon,Haotian Zhang,Ming Wang,Weiyong Yang,Xinshen Wei,Zhuo Lv & 2023

6- Social network malicious insider detection using time-based trust evaluation T Nathezhtha, D Sangeetha, V Vaidehi
2023

7- ACGAN: Adaptive Loss-Guided GAN-CNN Framework for Semantically Coherent Synthetic Insider Threat Data
Generation A Slim, M Malah 2025